Preliminary Examinations
Higher 1

TEMASEK JUNIOR COLLEGE

CANDIDATE NAME					
CIVICS GROUP			/		
CENTER NUMBER	S				

INDEX NUMBER | | | | |

CHEMISTRY 8872/02

Paper 2 **14^{th} September 2011**

Candidates answer Section A on the Question Paper. **2 hours**

Additional Materials: Answer Paper
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Civics Group number and candidate name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use		
Paper 1		/30
Paper 2	A1	/17
	A2	/11
	A3	/12
	B	/20
	B	/20
Total		/110

This document consists of **16** printed pages.

Section A

Answer **all** questions in this section in the spaces provided.

For Examiner's use

1 This question is about aluminium, silicon, phosphorus and sulfur.

(a) Describe the variations in melting points of the elements aluminium to sulfur and explain the variations in terms of their structures and bonding. **[3]**

Aluminium, silicon and phosphorus react with chlorine to form the chlorides of formulae $AlCl_3$, $SiCl_4$, PCl_3 / PCl_5 and S_2Cl_2 respectively.

(b) Draw the dot-and-cross diagrams for $AlCl_3$ and PCl_3, state the shape and suggest a value for each of the bond angles. **[3]**

$AlCl_3$	PCl_3
Shape:	Shape:
Bond angle:	Bond angle:

For Examiner's use

(c) State and explain the pH and the colour observed when some universal indicator is added to the solution formed from $AlCl_3$ dissolving in water. Write equations where appropriate. **[3]**

(d) When liquid $SiCl_4$ is added to water, a white solid of SiO_2 is immediately formed. Explain why $SiCl_4$ can be hydrolysed by water to form SiO_2. **[1]**

For Examiner's use

(e) The following table shows three compounds HCN, HCHO and $SiCl_4$. From their structures shown, deduce the number of σ bonds around central atom and types of hybridisation in the boxes below. **[2]**

	HCN	HCHO	$SiCl_4$
Number of σ bonds around central atom			
Type of hybridisation			

(f) Sulfur and chlorine can be reacted together to form disulfur dichloride, S_2Cl_2. Disulfur dichloride, S_2Cl_2, is decomposed by water forming sulfur and a mixture of hydrochloric acid and sulfurous acid.

When 1.35 g of S_2Cl_2 is reacted with an excess of water, 0.48 g of sulfur, S, is produced.

(i) What is the amount, in moles, of S_2Cl_2 present in 1.35 g?

(ii) What is the amount of S produced from 1.0 mol of S_2Cl_2?

For Examiner's use

(iii) Construct a balanced equation for the reaction of S_2Cl_2 with water.

....

(iv) State and explain the type of reaction in **(f)(iii)**.

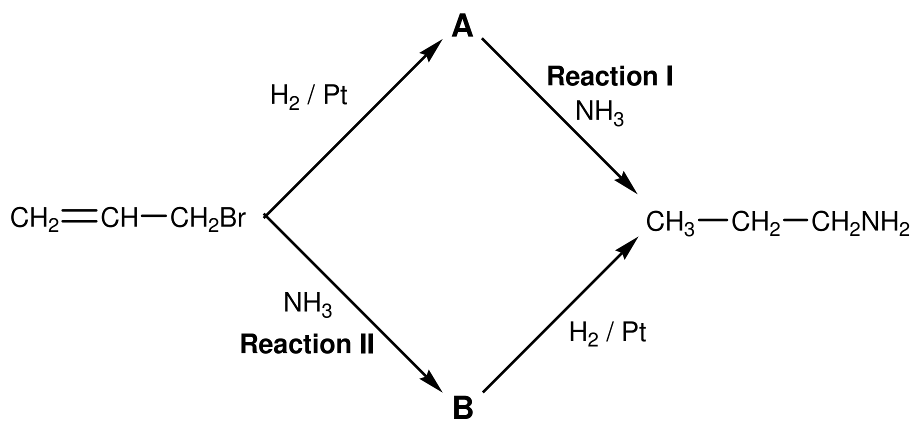
....

[5]

[Total: 17]

For Examiner's use

2 3-bromopropene is an alkylating agent used in the synthesis of polymers, pharmaceuticals and other organic compounds. 3-bromopropene is a clear liquid with an intense, acrid and persistent smell. 3-bromopropene can be used in the formation of 1-aminopropane though the following two processes.



(a) Give the structural formulae of the compounds **A** and **B**. **[2]**

A: …………………………………..

B: …………………………………

(b) Explain whether 3-bromopropene can show geometric isomerism. **[2]**

For Examiner's use

(c) **(i)** To investigate the kinetics of **Reaction I**, the time taken for a small constant amount of 1-aminopropane was measured when various concentrations of **A** and NH_3 were reacted. The results are shown in the table below:

Experiment	**[A] / mol dm^{-3}**	**[NH_3] / mol dm^{-3}**	**Time for formation of $CH_3CH_2CH_2NH_2$ / min**
1	0.40	4.00	18
2	0.20	4.00	36
3	0.40	6.00	12

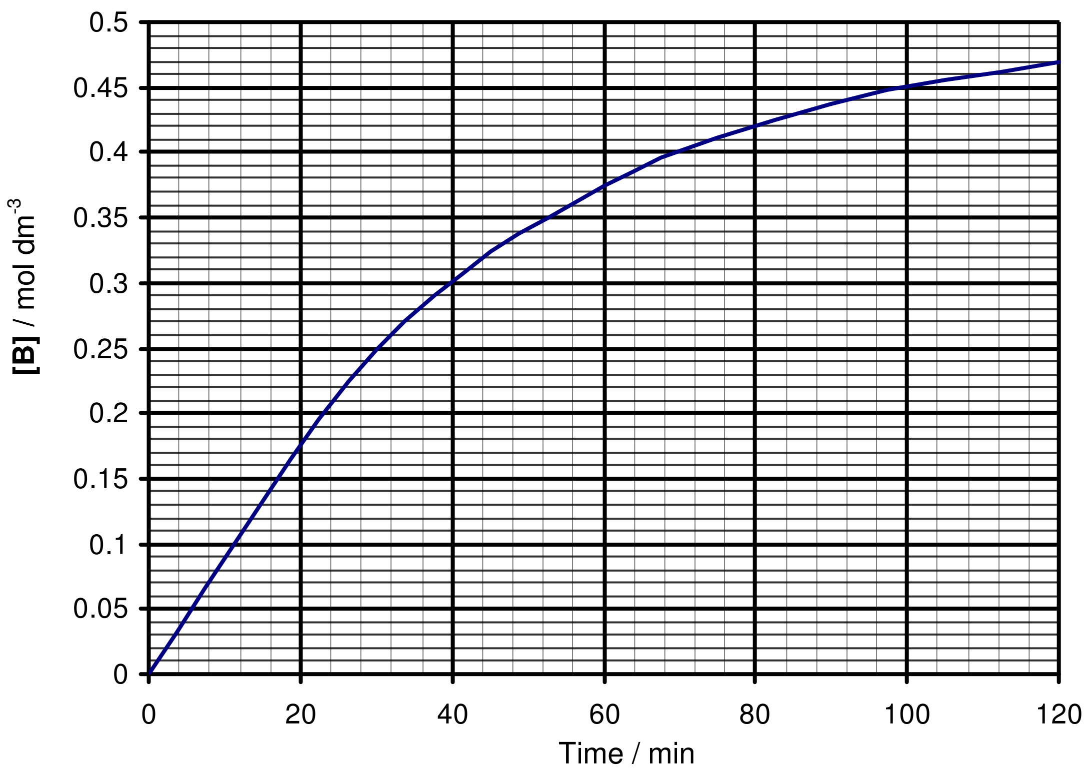
Determine the order of reaction with respect to **A** and NH_3 showing how you arrive at your answers.

(ii) Construct the rate equation for the reaction between **A** and NH_3.

[3]

For Examiner's use

(d) **(i)** To investigate the kinetics of **Reaction II**, the concentration of **B** produced with time was obtained plotted on a graph shown below. The initial concentrations of 3-bromopropene and NH_3 were 0.50 mol dm^{-3} and 2.50 mol dm^{-3} respectively. When the reaction is complete, the concentration of **B** is 0.50 mol dm^{-3}.



When the initial concentration of NH_3 was doubled, the same graph was obtained.

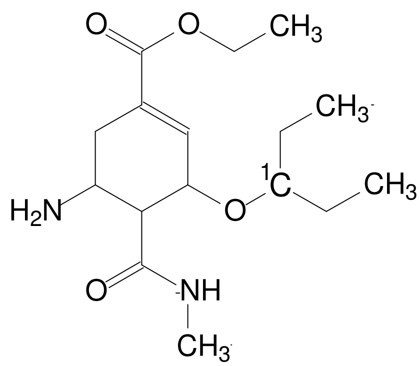
Using the graph and information above, deduce the orders of reaction with respect to NH_3 and 3-bromopropene.

(ii) Determine the rate constant for the reaction, stating the units.

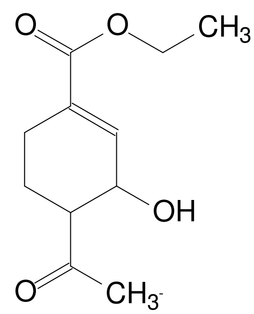
[4]
[Total: 11]

[Turn over

3 **Oseltamivir** (Tamiflu) is an antiviral drug that slows the spread of non-resistant strains of the influenza virus between cells in the body. It blocks the action of a viral enzyme called neuraminidase and has since been indicated for the treatment of H5N1 and H1N1 infection. The standard adult dosage is 75 mg twice daily. Compound **C** is a derivative of Oseltamivir that maybe investigated for antiviral activities.



Oseltamivir
M_r = 312.4



Compound C

(a) **(i)** A male adult patient has been put on a 1-week oseltamivir treatment. Calculate the total number of moles of oseltamivir taken by this patient over the period of treatment.

(ii) State the shape and bond angle about C_1 atom.

Shape: --

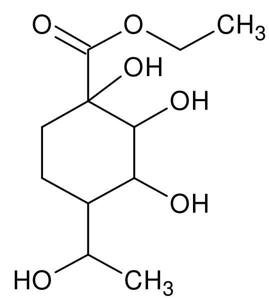
Bond angle: --

[2]

For Examiner's use

(b) **(i)** Name the functional groups present in Compound **C**.

(ii) Compound **D** can be synthesised from **C** using a 2-step reaction. State the reagents and conditions for both reactions. Name the types of reactions involved.



Compound D

Step I:

Reagent: --

Condition: --

Type of reaction: --

Step II:

Reagent: --

Condition: --

Type of reaction: --

[6]

For Examiner's use

(c) **(i)** Draw the products formed when Compound **C** is reacted with hot alkaline iodine.

(ii) Describe a chemical test to distinguish **C** from **D.**

....

....

....

....

....

[4]
[Total: 12]

Section B

Answer **two** of the three questions in this section on separate paper

4 **(a)** Concern over the ever-increasing use of fossil fuels has led to many suggestions for alternative sources of energy. One of these, suggested by Professor George Olah, winner of a Nobel Prize in chemistry, is to use methanol, CH_3OH, which can be manufactured in a number of ways.

One of the ways to manufacture methanol is from *synthesis gas*, a mixture of CO, CO_2 and H_2. The CO is reacted with H_2 to form methanol, CH_3OH.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \quad \Delta H = -91 \text{ kJ mol}^{-1}$$

(i) 0.750 mol each of CO and H_2 is injected into a 2 dm^3 vessel at 400 °C and allowed to reach equilibrium. 0.200 mol of CH_3OH was present at equilibrium.

Calculate the equilibrium constant, K_c, for the reaction.

(ii) The experiment was repeated using a vessel of a different volume and it was found that the equilibrium yield of CH_3OH was higher.

Deduce if the vessel used was larger or smaller.

(iii) This method of synthesising methanol can be catalysed by a mixture of copper and zirconia oxide.

Suggest and explain the effect on the equilibrium composition if another catalyst, MK-121, was used instead.

[6]

(b) **(i)** A mixture of ethane-1,2-diol and water is used as an antifreeze and is found to freeze when temperatures drop below -45 °C.

Explain, in terms of structure and bonding, why ethane-1,2-diol is miscible with water.

(ii) During winter, water freezes when the temperature drops below 0 °C and a layer of ice is found at the top of lakes as ice is less dense than water. With the aid of a diagram, explain this phenomenon.

[4]

(c) A colour change is observed when methanol is oxidised by hot acidified potassium manganate(VII).

(i) Write a balanced equation for this reaction.

(ii) Explain why methanol has been oxidised.

[2]

(d) 0.00754 mol of compound **E,** C_xH_yO, was burnt in excess oxygen to produce 1.66 g of carbon dioxide and 0.54 g of water.

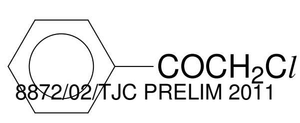
E decolourises aqueous bromine and reacts with PCl_5 giving off white fumes. Upon refluxing **E** with acidified potassium manganate(VII), a symmetrical product **F**, $C_5H_6O_5$, is formed. **F** does not give a red precipitate with Fehling's solution but a orange precipitate is observed with 2,4-dinitrophenylhydrazine. 1 mol of **F** also reacts with 1 mol of Na_2CO_3 with effervescence observed. **F** reacts with $NaBH_4$ to form **G**.

(i) Using the data above, verify that the molecular formula of **E** is C_5H_8O.

(ii) Deduce the structures of compounds **E**, **F** and **G**, explaining the chemistry of the reactions involved.

[8]
[Total: 20]

5 **(a)** Chloroacetophenone was formerly the most widely used tear gas, under the codename *CN*. It was used in warfare and in riot control.

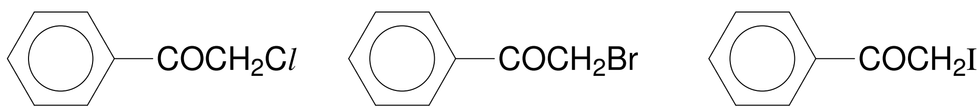


Chloroacetophenone

The efficiency of a tear gas is expressed by its 'intolerable concentration'. The intolerable concentration of the tear gas *CN* has been measured as 0.030 g m^{-3} of air.

How many moles of chloroacetophenone need to be sprayed into a room of volume 60 m^3 in order to achieve this concentration? **[2]**

(b) The compounds below differ in their physical properties and reactivities.



(i) Suggest and explain how the boiling points of these three compounds differ.

(ii) Suggest and explain how the C-X bond polarities in these three compounds differ.

(iii) Describe and explain how the reactivities of the three compounds differ towards reaction with aqueous sodium hydroxide.

(iv) Describe how you would distinguish the three compounds above, writing balanced equations for reactions mentioned.

[10]

(c) Upon irradiation of ultra-violet light, 2,4-dimethylpentane undergoes monochloro-substitution.

(i) Draw the structural formula of all the possible monochlorinated products.

(ii) Predict the mole ratio of the monochlorinated products in **(c)(i)**.

(iii) The hydrogen in an organic molecule can be classified as primary, secondary or tertiary hydrogen as illustrated below.

H R' R'

|	|	|
R-C-H	R-C-H	R-C-R''
|	|	|
H	H	H
Primary H	Secondary H	Tertiary H

You are provided with the following bond dissociation enthalpy data:

Bond	Bond dissociation enthalpy / kJ mol^{-1}
H-C*l*	+432
C*l*-C*l*	+243
$H\text{-}CH_2CH_3$	+420
$H\text{-}CH(CH_3)_2$	+401
$H\text{-}C(CH_3)_3$	+390
$Cl\text{-}CH_2CH_3$	+338
$Cl\text{-}CH(CH_3)_2$	+339
$Cl\text{-}C(CH_3)_3$	+330

Using the equation for the monochlorination of an alkane below, estimate the respective enthalpy changes of the monochlorination of primary, secondary and tertiary hydrogen atoms in 2,4-dimethylpentane.

$$RH + Cl\text{-}Cl \longrightarrow RCl + H\text{-}Cl$$

(iv) From your answer in **(c)(iii)**, suggest a factor that could have affected the yield of the monochlorinated products from 2,4-dimethylpentane. Explain your answer.

[8]
[Total: 20]

6 **(a)** Sodium hydroxide is often used for acid base titrations. A student prepared a standard solution of sodium hydroxide by dissolving 10.0 g of sodium hydroxide in 250 cm^3 of water. During the preparation, he found the solution warm to the touch.

(i) Calculate the concentration of the sodium hydroxide solution prepared and calculate its pH.

(ii) Given the information below, calculate the heat evolved when solid sodium hydroxide is dissolved in water.

$$NaOH(s) + aq \rightarrow NaOH(aq)$$

	ΔH_f / kJ mol^{-1}
NaOH(s)	-425
NaOH(aq)	-470

Hence, find the increase in temperature of the standard solution.

[6]

(b) Carboxylic acids are *weak Bronsted acids*.

(i) What do you understand by the term "*weak Bronsted acids*"?

The following table gives the pK_a values of three organic compounds.

Compound	**M_r**	**pK_a**	**Melting point / °C**
Propan-1-ol	60.0	16.0	-126
Ethanoic acid	60.0	4.79	17
Chloroethanoic acid	94.5	2.86	63

(ii) Arrange the organic compounds in increasing acidity. Explain your answer.

(iii) Explain, in terms of bonding and structure, the difference in melting point between propan-1-ol and ethanoic acid.

(iv) Suggest a 3-step synthesis of ethanol from propan-1-ol.

(v) Write a balanced equation for the reaction between ethanoic acid and magnesium metal.

[11]

(c) Mixture **J** is a *buffer* that is made up of 50 cm^3 of 1 mol dm^{-3} sodium hydroxide solution and 100 cm^3 of 1 mol dm^{-3} of ethanoic acid.

(i) What is meant by a *buffer*?

(ii) Write equations to show how mixture **J** acts as a buffer.

[3]
[Total: 20]